General Relativity — Final Examination
Answers

Question 1. *(2 points)*

Under a change of coordinates $x^\mu \to x'^\mu(x)$ the Riemann tensor transforms as a $(1,3)$ tensor:

$$R'^{\mu}{}_{\nu\rho\sigma} = \frac{\partial x'^\mu}{\partial x^\alpha}\,\frac{\partial x^\beta}{\partial x'^\nu}\,\frac{\partial x^\gamma}{\partial x'^\rho}\,\frac{\partial x^\delta}{\partial x'^\sigma}\;R^{\alpha}{}_{\beta\gamma\delta}\,. \tag{1}$$

(This follows because $R^{\mu}{}_{\nu\rho\sigma}V^\nu = [\nabla_\rho, \nabla_\sigma]V^\mu$ for a torsion-free connection. The object $\nabla_\rho\nabla_\sigma V^\mu$ is a $(1,2)$ tensor, so the commutator is a tensor. The derivative terms ∂V cancel in the commutator, so it is linear and algebraic in V^ν. By the quotient theorem $R^{\mu}{}_{\nu\rho\sigma}$ is then a tensor.)

Now contract the first and third indices, $R'_{\nu\sigma} = R'^{\mu}{}_{\nu\mu\sigma}$:

$$R'_{\nu\sigma} = \frac{\partial x'^\mu}{\partial x^\alpha}\,\frac{\partial x^\beta}{\partial x'^\nu}\,\frac{\partial x^\gamma}{\partial x'^\mu}\,\frac{\partial x^\delta}{\partial x'^\sigma}\;R^{\alpha}{}_{\beta\gamma\delta}\,. \tag{2}$$

The chain rule gives

$$\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x^\gamma}{\partial x'^\mu} = \frac{\partial x^\gamma}{\partial x^\alpha} = \delta^\gamma_\alpha\,, \tag{3}$$

and therefore

$$R'_{\nu\sigma} = \frac{\partial x^\beta}{\partial x'^\nu}\,\frac{\partial x^\delta}{\partial x'^\sigma}\;\delta^\gamma_\alpha R^{\alpha}{}_{\beta\gamma\delta} = \frac{\partial x^\beta}{\partial x'^\nu}\,\frac{\partial x^\delta}{\partial x'^\sigma}\;R^{\alpha}{}_{\beta\alpha\delta} = \frac{\partial x^\beta}{\partial x'^\nu}\,\frac{\partial x^\delta}{\partial x'^\sigma}\;R_{\beta\delta}\,. \tag{4}$$

This is the transformation law of a $(0,2)$ tensor, so the Ricci tensor is a tensor. The Jacobians of the contracted pair cancel because contraction of an upper with a lower index is coordinate independent.

Question 2. *(3 points)*

Write $\nabla_\mu V^\rho = \partial_\mu V^\rho + \Gamma_{\mu\nu}{}^\rho V^\nu$ (the argument below does not depend on which lower slot is the derivative index, because H is totally antisymmetric).

a) *Torsion.* The torsion is the antisymmetric part of the connection. The Levi-Civita part is symmetric in $\mu\nu$, so it drops out:

$$\hat{T}_{\mu\nu}{}^\rho = \hat{\Gamma}_{\mu\nu}{}^\rho - \hat{\Gamma}_{\nu\mu}{}^\rho = \frac{1}{12}g^{\rho\sigma}\left(H_{\mu\nu\sigma} - H_{\nu\mu\sigma}\right) = \frac{1}{6}H_{\mu\nu}{}^\rho\,. \tag{5}$$

This is non-zero whenever $H \neq 0$, so the connection is *not* torsion-free.

Metric compatibility.

$$\begin{aligned}\hat{\nabla}_\mu g_{\nu\rho} &= \partial_\mu g_{\nu\rho} - \hat{\Gamma}_{\mu\nu}{}^\sigma g_{\sigma\rho} - \hat{\Gamma}_{\mu\rho}{}^\sigma g_{\nu\sigma} \\ &= \underbrace{\partial_\mu g_{\nu\rho} - \Gamma_{\mu\nu}{}^\sigma g_{\sigma\rho} - \Gamma_{\mu\rho}{}^\sigma g_{\nu\sigma}}_{=\nabla_\mu g_{\nu\rho}=0} - \frac{1}{12}\left(H_{\mu\nu\rho} + H_{\mu\rho\nu}\right) = 0\,, \end{aligned} \tag{6}$$

since $H_{\mu\nu\rho} = -H_{\mu\rho\nu}$. So the connection *is* metric compatible.

b) The geodesic (autoparallel) equation for $\hat{\Gamma}$ is

$$\frac{d^2x^\rho}{d\tau^2} + \hat{\Gamma}_{\mu\nu}{}^\rho \frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0\,. \tag{7}$$

Substituting the definition,

$$\frac{d^2x^\rho}{d\tau^2} + \Gamma_{\mu\nu}{}^\rho \dot{x}^\mu \dot{x}^\nu + \frac{1}{12} g^{\rho\sigma} H_{\mu\nu\sigma} \dot{x}^\mu \dot{x}^\nu = 0\,. \tag{8}$$

The last term contracts the antisymmetric $H_{\mu\nu\sigma}$ with the symmetric $\dot{x}^\mu\dot{x}^\nu$, so it vanishes identically. The equation is then the Levi-Civita geodesic equation, so the geodesics are the same.

Question 3. *(4 points)*

a) The metric components are singular at three places:

- $\theta = 0, \pi$, where $\sin\theta = 0$. This is just the usual coordinate singularity of spherical polar coordinates (as at the poles of a sphere), and is harmless.
- $r = 2GM$, where $g_{tt} \to 0$ and $g_{rr} \to \infty$. This is the event horizon. It is a *coordinate singularity*: the Schwarzschild t coordinate is badly adapted there (an infalling object takes $t \to \infty$ to reach it). No curvature scalar diverges, for example $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 48G^2M^2/r^6$ is finite at $r = 2GM$. Tidal forces there are finite and, for a large black hole, small.
- $r = 0$. Here $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 48G^2M^2/r^6 \to \infty$, so this is a genuine *curvature singularity*, where tidal forces diverge and geodesics end. This is the one to be concerned about. A radially infalling observer starting at rest at $r = 2GM$ reaches it in finite proper time πGM, and is torn apart by infinite tidal forces.

b) Let $r_* = r + 2GM\ln|r/2GM - 1|$, so $v = t + r_*$. Then

$$\frac{dr_*}{dr} = 1 + \frac{2GM}{r - 2GM} = \frac{r}{r - 2GM} = \frac{1}{f}\,, \tag{9}$$

so $dv = dt + dr/f$, i.e. $dt = dv - dr/f$. Substituting,

$$-f\,dt^2 + \frac{dr^2}{f} = -f\left(dv^2 - \frac{2\,dv\,dr}{f} + \frac{dr^2}{f^2}\right) + \frac{dr^2}{f} = -f\,dv^2 + 2\,dv\,dr\,, \tag{10}$$

and therefore

$$ds^2 = -f(r)\,dv^2 + 2\,dv\,dr + r^2 d\theta^2 + r^2\sin^2\theta\,d\phi^2\,. \tag{11}$$

All components are smooth at $r = 2GM$. The (v, r) block $\begin{pmatrix} -f & 1 \\ 1 & 0 \end{pmatrix}$ has determinant -1, so $\det g = -r^4\sin^2\theta \neq 0$ and the metric is invertible and regular across $r = 2GM$. It extends smoothly to $r < 2GM$, so the singularity at $r = 2GM$ was an artefact of the coordinates.

The only remaining divergence is at $r = 0$. The radial null curves satisfy $ds^2 = 0$, so either $dv = 0$ (ingoing rays, $v = \text{const.}$) or $dv/dr = 2/f$ (outgoing rays). The ingoing rays cross $r = 2GM$ smoothly. For $r < 2GM$ we have $f < 0$, so $dr/dv = f/2 < 0$ on both families. Both light-cone edges point towards decreasing r, so every future-directed path reaches $r = 0$, and nothing can come back out. The infalling observer crosses the horizon with nothing special happening locally, and the only genuinely dangerous place is $r = 0$. This supports the answer to part a).

Question 4. *(5 points)*

a) The metric has the FRW form $ds^2 = -dt^2 + a(t)^2 d\Omega_3^2$ with spatial slices that are three-spheres, so it is homogeneous and isotropic as is observed on large scales (a $k = +1$ FRW universe). The *scale factor* is

$$a(t) = L\cosh(t/L)\,. \tag{12}$$

It solves the Friedmann equation with positive cosmological constant $\Lambda = 3/L^2$ and no matter:

$$\left(\frac{\dot a}{a}\right)^2 + \frac{1}{a^2} = \frac{\tanh^2(t/L)}{L^2} + \frac{1}{L^2\cosh^2(t/L)} = \frac{1}{L^2} = \frac{\Lambda}{3}\,. \tag{13}$$

At late times $a \sim \frac{L}{2}e^{t/L}$, i.e. exponential, accelerating expansion with Hubble constant $H = 1/L$. This matches the observation that our universe is expanding at an accelerating rate, dominated by dark energy / a cosmological constant. So it is a viable candidate (at least for the late universe).

b) Null curves satisfy $ds^2 = 0$. If $g_{\mu\nu} = \Omega^2\bar g_{\mu\nu}$ with Ω smooth and non-vanishing, then $g_{\mu\nu}\dot x^\mu\dot x^\nu = \Omega^2\bar g_{\mu\nu}\dot x^\mu\dot x^\nu$ has the same sign as $\bar g_{\mu\nu}\dot x^\mu\dot x^\nu$, because $\Omega^2 > 0$. A curve is timelike, null or spacelike in g if and only if it is so in $\bar g$. The light cones at every point

are therefore identical, and so are the sets of causal curves and the causal past and future of every point. The causal structure is determined by the light cones alone, so it is the same. (Proper times and geodesics of massive particles do change, but only the light-cone structure matters for causality.)

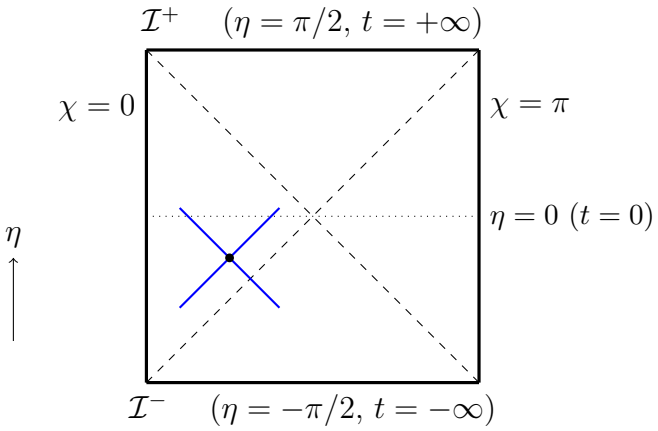
c) Let $\cosh(t/L) = 1/\cos\eta$. Then $\sinh(t/L) = \tan\eta$ (for $\eta \in (-\pi/2, \pi/2)$, with the sign matching t). Differentiating,

$$\frac{1}{L}\sinh(t/L)\,dt = \frac{\sin\eta}{\cos^2\eta}\,d\eta \;\Rightarrow\; \frac{\tan\eta}{L}\,dt = \frac{\sin\eta}{\cos^2\eta}\,d\eta \;\Rightarrow\; dt = \frac{L\,d\eta}{\cos\eta}\,. \tag{14}$$

Also $L^2\cosh^2(t/L) = L^2/\cos^2\eta$. The metric becomes

$$ds^2 = \frac{L^2}{\cos^2\eta}\left(-d\eta^2 + d\chi^2 + \sin^2\chi\,d\Omega_2^2\right). \tag{15}$$

This is conformal, with $\Omega = L/\cos\eta$, to the static Einstein universe $\mathbb{R}\times S^3$. As $t\to\pm\infty$, $\eta\to\pm\pi/2$, so the range is $\eta\in(-\pi/2,\pi/2)$ and the metric is conformal to a *finite* piece of $\mathbb{R}\times S^3$. By part b) it has the same causal structure. Radial null rays are $\eta = \pm\chi$+const., i.e. lines at 45°. The Penrose diagram is therefore a square, of width π in χ and height π in η:



The vertical sides are the poles $\chi = 0$ and $\chi = \pi$ of the three-sphere. They are not boundaries, just the centres of the polar coordinates (worldlines of observers sitting at the two poles). The top and bottom edges are spacelike infinity $\mathcal{I}^\pm$. The 45° light rays (blue) from a sample event show its light cone. The dashed diagonals show the horizons of an observer at a pole: there is a particle horizon (the observer cannot see events beyond it) and an event horizon (the observer will never see events beyond it). The fact that the diagram is a square, with $\mathcal{I}^+$ and $\mathcal{I}^-$ spacelike, shows that two antipodal observers can never exchange light signals.

Question 5. *(6 points)*

a) Under $x^\mu \to x^\mu + \xi^\mu$, $h_{\mu\nu} \to h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$ and

$$\bar{h}_{\mu\nu} \to \bar{h}_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu - \eta_{\mu\nu} \partial \cdot \xi \,, \qquad \partial^\mu \bar{h}_{\mu\nu} \to \partial^\mu \bar{h}_{\mu\nu} + \partial^\rho \partial_\rho \xi_\nu \,. \tag{16}$$

We can always choose ξ_ν solving $\Box \xi_\nu = -\partial^\mu \bar{h}_{\mu\nu}$, so the *Lorenz (de Donder) gauge* $\partial^\mu \bar{h}_{\mu\nu} = 0$ can be reached. In this gauge the second term, $\partial^\rho \partial_{(\mu} \bar{h}_{\nu)\rho} = \frac{1}{2} \partial_\mu (\partial^\rho \bar{h}_{\nu\rho}) + \frac{1}{2} \partial_\nu (\partial^\rho \bar{h}_{\mu\rho})$, and the third term, $\propto \partial^\rho \partial^\sigma \bar{h}_{\rho\sigma}$, both vanish. The linearised Einstein equation reduces to

$$\Box \, \bar{h}_{\mu\nu} = -16\pi G \, T_{\mu\nu} \,, \qquad \Box = -\partial_t^2 + \nabla^2 \,. \tag{17}$$

In vacuum, $T_{\mu\nu} = 0$, this is the wave equation $(-\partial_t^2 + \nabla^2) \bar{h}_{\mu\nu} = 0$. It has plane-wave solutions $\bar{h}_{\mu\nu} = \mathrm{Re}\,[A_{\mu\nu} e^{i k_\mu x^\mu}]$ with $k^2 = 0$, i.e. disturbances of the metric propagating at the speed of light: gravitational waves. Sources $T_{\mu\nu}$ generate them.

b) Test particles initially at rest remain at fixed coordinate positions for this perturbation. To linear order, with $u^\mu \approx (1, 0, 0, 0)$,

$$\frac{d^2 x^i}{d\tau^2} = -\Gamma^i_{00} = -\tfrac{1}{2} \left(2\partial_0 h_{0i} - \partial_i h_{00} \right) = 0 \,, \tag{18}$$

since $h_{0\mu} = 0$. The particles therefore stay at $x^i = (0, 0, 0)$ and $(L_x, 0, 0)$, and only the metric changes.

The proper distance along the x-axis at fixed t (with $dy = dz = 0$) is determined by $ds^2 = (1 + h_+(t - z))\, dx^2$. Along the path $z = 0$, $h_+(t - z) = h_+(t)$, independent of x, so

$$\Delta s = \int_0^{L_x} \sqrt{1 + h_+(t)}\, dx = L_x \sqrt{1 + h_+(t)} \approx L_x \left(1 + \tfrac{1}{2} h_+(t) \right) , \tag{19}$$

using $|h_+| \ll 1$ (linearised theory) to expand the square root.

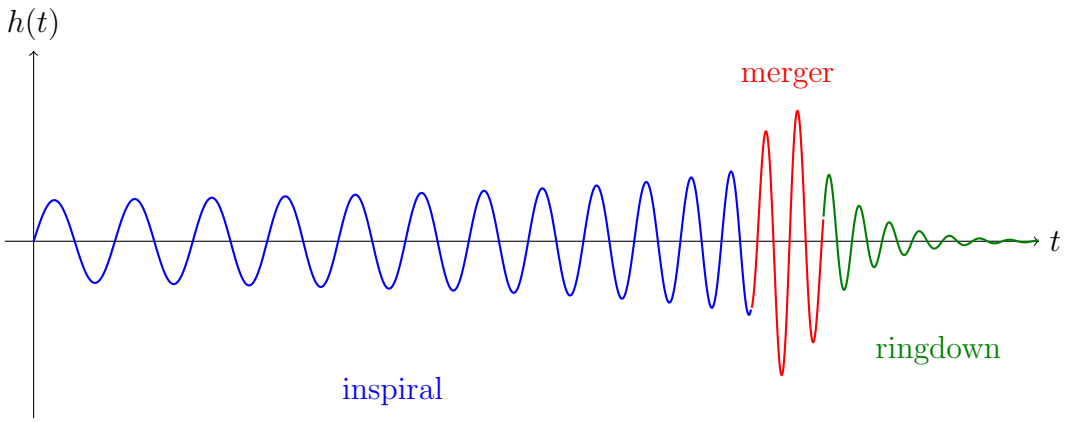
For the third particle at $(0, L_y, 0)$ we have $g_{yy} = 1 - h_+$, so $ds^2 = (1 - h_+(t))\, dy^2$ along the y-axis, and

$$\Delta s = \int_0^{L_y} \sqrt{1 - h_+(t)}\, dy = L_y \sqrt{1 - h_+(t)} \approx L_y \left(1 - \tfrac{1}{2} h_+(t) \right) . \tag{20}$$

The wave stretches distances along x while squeezing them along y, and vice versa half a period later: the quadrupolar + pattern.

c) *Physics of the binary.* Two bodies in orbit have a time-varying mass quadrupole, so they radiate gravitational waves, which carry away energy and angular momentum. The orbit therefore shrinks and the orbital frequency increases. There are three stages:

- *Inspiral.* The separation shrinks slowly and the orbit is quasi-circular. The gravitational-wave frequency is twice the orbital frequency. Both the frequency and amplitude increase with time, $f \propto (t_c - t)^{-3/8}$ and $h \propto f^{2/3}$ (a "chirp"), with the rate set by the chirp mass. The signal is a sinusoid with rising pitch and amplitude.
- *Merger.* When the separation reaches roughly the innermost stable circular orbit (a few GM/c^2) the bodies plunge together and merge into a single object. The fields are strong and non-linear (this needs numerical relativity). The amplitude peaks, and the frequency is highest, around here.
- *Ringdown.* The newly formed, distorted black hole settles to a stationary Kerr black hole by emitting gravitational waves at its quasi-normal-mode frequencies. The signal is an exponentially damped sinusoid, $h \sim e^{-t/\tau}\cos(\omega t)$, whose frequency ω and damping time τ are fixed by the final mass and spin alone (the no-hair theorem).



(Schematic strain: amplitude and frequency increase through the inspiral, peak at merger, then decay in the ringdown.)

Detection. An interferometer such as LIGO is a Michelson interferometer with two perpendicular arms (one along x, one along y) of length $L \approx 4$ km, with a test mass (mirror) at the end of each arm. By part b), a +-polarised wave travelling along z changes the arm lengths as $L_x \to L(1 + \frac{1}{2}h_+)$ and $L_y \to L(1 - \frac{1}{2}h_+)$. The arm-length difference is $\Delta L = L_x - L_y \approx h_+ L$, so the strain is $h_+ = \Delta L/L$. A laser beam is split into the two arms and recombined after reflecting from the mirrors. The interferometer is set up so that, with no wave, the two beams interfere destructively at the photodetector. The differential change in the round-trip light travel time changes the relative phase, so a signal appears in the detector output that follows $h_+(t)$. For $h \sim 10^{-21}$ and $L = 4$ km this means $\Delta L \sim 10^{-18}$ m, which is why long arms (with Fabry–Pérot cavities that fold the light path many times) are needed.

Matching the signal features to the physics: the rising frequency and amplitude of the chirp measure the chirp mass (the shrinking orbit); the peak at merger gives the

total mass and the size of the system when it merges; the ringdown frequency and decay time give the mass and spin of the final black hole; and the amplitude gives the distance to the source.